# SHEET # 3

3.1 Describe the convolution and correlation operations.

3.2 Define the following:

(a)Cross-correlation between two sequences $x_1(n)$ and $x_2(n)$ having lengths $N_1$ and $N_2$ respectively.

(b) Correlation theorem.

(c) Correlation coefficients.

3.3 Use the correlation theorem to calculate the circular correlation of the data sequences {1,1,0,1} and {1,0,0,1}. Plot the correlation function against lag number, j.

3.4 Apply the correlation theorem to calculate the linear correlation of the data sequences {1,1,0,1} and {1,0,0,1}. Plot the correlation function against lag numbers and compare the results with the solution to prob.2.7, explain any differences.

3.5 Given the two sets of data {1.5, 2.0, 1.5, 2.0, 2.5} and {0, 0.33, 0.67, 1.0} calculate the following:

(a) Autocorrelation of the two sequences.

(b) Cross-correlation using direct method.

(c) Cross-correlation using correlation theorem.

3.6 Determine the output of an electrical system of impulse response function { 0, 0.899, 0.990, 0.991, 1} when the input{ 0, 2.5, 5.0, 0 } volts is applied

(a) By direct convolution.

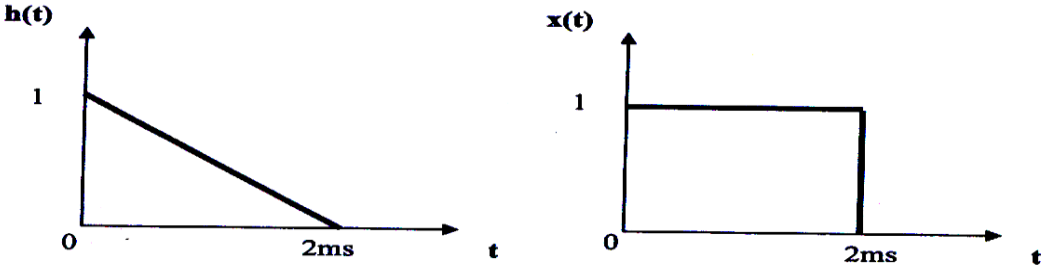
(b) By applying the convolution theorem.

3.7 A unit amplitude pulse of width 2ms (Fig.3.15b) is applied to a circuit which has an impulse response shown in Fig.3.15a. Determine the output waveform:

(a) Analytically.

(b) By sampling h(t) every 0.25ms and representing the pulse by an impulse function located at t = 0s.

(c) Compare your results in part (a) and (b).




(a) (b)

Fig. 3.15

3.8 For Prob. (3.7), find the cross-correlation between the given waveforms using direct method assume the sampling interval $T_S = 0.2\,msec$.

3.9 For the given waveform, Fig. 3.16, compute the convolution with itself y(t). Then assume the sampling interval $T_S = 0.2\ msec$ find y(n). Compare y(t) and y(n).

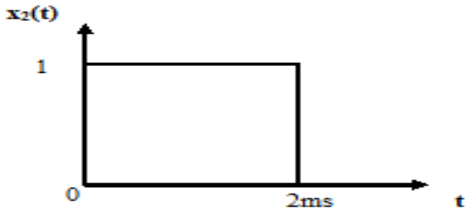


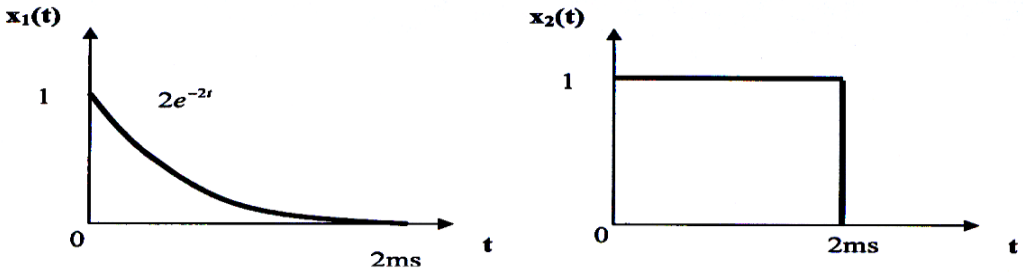

Fig. 3.16

3.10 Figure 3.17 show two functions x1(t) and x2(t). Evaluate the following:

(a) Their convolution, $x_3(t)$ , numerically, taking sampled values at t = 0 , 0.5,1,1.5 and 2ms.

(b) x3(t) analytically.

(c) Sketch the function x3(t) and give reasons for any difference between them.




(a) (b)

Fig. 3.17

# Assignment # 3

A rectangular pulse of height 5V and width $10\mu$ sec shown in Fig. 3.18, is applied to the input of a system with the response function h(t) given by:

$$h(t) = \begin{cases} 0.1(1-e^{\frac{-t}{1.09\times10^{-6}}}) & 0 \le t \le 10\mu\text{sec} \\ 0 & 10\mu\text{sec} < t < \infty \end{cases}$$

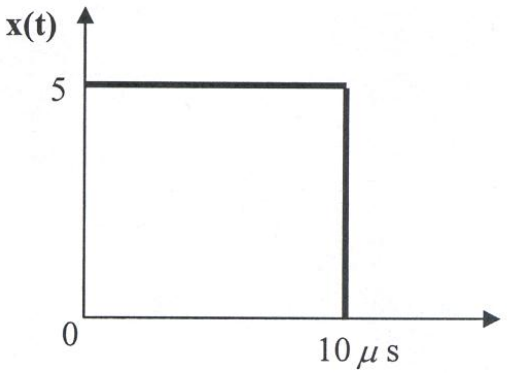



Fig. 3.18

Determine the system output:

(a) Analytically.

(b) By sampling h (t) every $1\mu\text{sec}$ and representing the pulse by impulse function located at t = 0sec.

(c) Compare your results in part (a) and (b).